Predicting Plasticity in Deep Continual Learning: A Theoretical Perspective

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Abstract

Deep continual learning requires models to adapt to new tasks without retraining from scratch. However, neural networks can lose their ability to adapt to new tasks after training on previous ones, a phenomenon known as loss of plasticity. There have been several explanations and diagnostics proposed for plasticity loss. Motivated by the philosophy that "all models are wrong, but some are useful", we ask: can existing diagnostics predict a neural network's plasticity? In this work, we take a practical view to interpret plasticity as trainability, i.e., a neural network's future optimization gain on a target task. We first take a theoretical approach, showing, by constructing a few counterexamples, that some widely adopted diagnostics of plasticity, including representation rank and neural tangent kernel rank, can fail to predict the loss of trainability in both regression and classification settings. We instead propose a novel metric, called optimization readiness, which combines gradient strength and gradient reliability. We prove that optimization readiness lower bounds one-step optimization gain under standard smoothness assumptions, providing a theoretical guarantee for its predictive power. Empirically, we show that across commonly used deep continual learning settings, such as Slowly-Changing Regression and Permuted MNIST, optimization readiness more reliably ranks checkpoints by trainability than prior diagnostics, even with substantially fewer samples.

1 Introduction

Deep continual learning describes a machine learning setting in which a neural network learns from a sequence of tasks, where the learner needs to adapt sequentially while retaining useful knowledge from earlier experience [Aljundi, 2019, Parisi et al., 2019]. Such continual adaptation is possible only if the neural network model remains trainable, i.e., it can achieve meaningful improvement on the new task via gradient-based optimization. Unfortunately, continual adaptation cannot be achieved simply by continuing gradient-based training on newly arriving data, because neural networks can suffer from loss of plasticity [Dohare et al., 2024]. Loss of plasticity is the phenomenon in which a previously trained model loses its ability to adapt to new tasks, often performing worse than a randomly initialized network after the same amount of training on the new task. In severe cases,

Preprint.

optimization may stagnate, leaving the learning curve flat and posing a serious challenge to continual adaptation [Abbas et al., 2023].

Several diagnostics and mechanisms are proposed for explaining plasticity loss. For instance, the effective rank of the representations [Dohare et al., 2024], the rank of the empirical neural tangent kernel [Lyle et al., 2025, Tang et al., 2025], the directions of curvature [Lewandowski et al., 2024], and the proportion of dead/dormant neurons [Sokar et al., 2023, Ma et al., 2024]. Motivated by Box's philosophy that "all models are wrong, but some are useful," [Box, 1976, Box and Draper, 1986] we ask: can existing diagnostics predict a neural network's plasticity? To study this question, we first need to operationalize plasticity. Unfortunately, plasticity itself is operationalized differently across the literature. Dohare et al. [2024], Kumar et al. [2025] evaluate plasticity through learning speed. Wang et al. [2026] evaluate plasticity through final performance after a fixed training budget. Abbas et al. [2023], Lyle et al. [2023], Nikishin et al. [2023] evaluate plasticity through performance relative to a randomly initialized model. A consensus we perceive is that a network is plastic if additional gradient-based optimization produces a meaningful reduction in loss. Here, by meaningful reduction, we mean a reduction that is significant relative to the initial loss. We refer to this perceived consensus as trainability, and we view it as a concrete aspect of plasticity in this work. We will later mathematically formalize trainability as the k-step gain, the relative loss reduction achieved after k gradient steps on the target task, and use it as the target we wish to predict from diagnostic metrics.

Recent empirical evidence suggests that some commonly used diagnostics are unreliable indicators of plasticity [Lyle et al., 2023, Lewandowski et al., 2024, Baveja et al., 2025]. However, existing falsification is largely empirical, and there remains a limited theoretical understanding of why such metrics can fail.

In this work, we show that several structural diagnostics can be misleading: they may assign a favorable value to a neural network checkpoint even when gradient descent cannot make progress on the relevant task. In particular, our theoretical counterexamples focus on label-agnostic structural metrics, such as representation rank and eNTK rank, which depend on the input distribution and checkpoint but not directly on task labels. These negative results suggest that favorable structural diagnostic values alone are insufficient for predicting future optimization progress. We therefore propose an optimization-derived metric that is directly motivated by the one-step descent dynamics of stochastic gradient descent.

Specifically, our paper makes the following contributions:

1. We prove by construction of counterexamples that some widely-adopted diagnostics of plasticity fail to predict the loss of trainability in both regression and binary classification settings. In particular, we show that label-agnostic metrics, such as the effective rank, the 99% energy rank of the representation matrix, and the eNTK, can fail even in simple constructions. These results show that favorable structural diagnostic values alone do not guarantee future optimization progress.

2. Motivated by these negative results, we propose Optimization Readiness (OR), a lightweight optimization-derived metric designed to predict trainability. We prove that OR lower-bounds the one-step optimization gain under standard smoothness assumptions, providing a theoretical guarantee of its predictive power.

3. Empirically, we evaluate OR on Slowly-Changing Regression and Permuted MNIST, two standard testbeds of loss of plasticity [Dohare et al., 2024], and demonstrate that it accurately predicts the relative ordering of checkpoints by their trainability. In our ablation study, we additionally show that OR is data-efficient, achieving high pairwise ranking accuracy even with 10%, 1%, and 0.1% of the validation data in Slowly-Changing Regression and 10% and 1% of the Permuted MNIST validation set.

2 Preliminaries and Problem Setup

We study whether a diagnostic computed at a checkpoint can predict its *trainability* on a newly observed supervised task. A continual supervised-learning problem is a sequence of tasks $\mathcal{T}_1, \mathcal{T}_2, \ldots$, where each task $\mathcal{T} = (\mathcal{X}, \mathcal{Y}, P_{\mathcal{X}\mathcal{Y}})$ consists of an input space, a label space, and a joint distribution over $\mathcal{X} \times \mathcal{Y}$. Tasks arrive sequentially, and samples within each task are drawn i.i.d. from the corresponding distribution. A learner is a neural network $f_\theta : \mathcal{X} \to \mathcal{Z}$, and all tasks share a pointwise loss $\ell : \mathcal{Z} \times \mathcal{Y} \to \mathbb{R}_{\geq 0}$. The population loss on task $\mathcal{T}$ is $\mathcal{L}(\theta; \mathcal{T}) \doteq \mathbb{E}_{(x,y)\sim P_{\mathcal{XY}}}[\ell(f_\theta(x), y)]$.

Trainability target. For a checkpoint θ, our target quantity is the relative loss reduction obtained by continuing gradient-based optimization on $\mathcal{T}$. Let $g(\theta; \mathcal{T}) \doteq \nabla_\theta \mathcal{L}(\theta; \mathcal{T})$ denote the population gradient. For a mini-batch $\mathcal{B} \doteq \{(x_i, y_i)\}_{i=1}^m$, define $\widehat{\mathcal{L}}_{\mathcal{B}}(\theta; \mathcal{T}) \doteq \frac{1}{m}\sum_{i=1}^m \ell(f_\theta(x_i), y_i)$ as the empirical loss and $\widehat{g}_{\mathcal{B}}(\theta; \mathcal{T}) \doteq \nabla_\theta \widehat{\mathcal{L}}_{\mathcal{B}}(\theta; \mathcal{T})$. Starting from $\theta_0 = \theta$, stochastic gradient descent with learning rate $\eta > 0$ follows

$$\theta_{t+1} = \theta_t - \eta \widehat{g}_{\mathcal{B}_t}(\theta_t; \mathcal{T}), \tag{1}$$

for $t = 0, \ldots, k-1$, where $\mathcal{B}_0, \ldots, \mathcal{B}_{k-1}$ are sampled i.i.d. from $P_{\mathcal{XY}}$. We define the k-step gain as

Definition 2.1 (mini-batch k-step gain)**.**

$$\mathcal{G}^{(k)}(\theta; \mathcal{T}) \doteq \begin{cases} \dfrac{\mathcal{L}(\theta; \mathcal{T}) - \mathbb{E}_{\mathcal{B}_0,\ldots,\mathcal{B}_{k-1}}[\mathcal{L}(\theta_k; \mathcal{T})]}{\mathcal{L}(\theta; \mathcal{T})}, & \mathcal{L}(\theta; \mathcal{T}) > 0, \\ 0, & \mathcal{L}(\theta; \mathcal{T}) = 0. \end{cases}$$

Thus, $\mathcal{G}^{(k)}$ measures the expected relative improvement after k gradient steps, and we use it as an operational proxy for plasticity. When the neural network loses its plasticity, we expect the k-step gain to be small across different k's. Normalization in $\mathcal{G}^{(k)}$ is important because the absolute reduction in loss can be misleading across checkpoints with different initial losses. For example, a checkpoint close to convergence may have little room for absolute improvement, and therefore may exhibit a smaller absolute loss reduction than a randomly initialized checkpoint. This does not necessarily mean that it is less trainable — after the same number of optimization steps, it may reduce its remaining loss by a comparable, or even larger, proportion. Normalizing by the initial loss, therefore, makes the gain comparable across checkpoints at different stages of optimization.

Matrix-rank diagnostics. Several existing plasticity diagnostics are based on the spectrum of representations, neural tangent kernels (NTK, Jacot et al. [2018]), or curvature matrices. For a matrix $Z \in \mathbb{R}^{m\times n}$, let $\sigma_1 \geq \cdots \geq \sigma_q \geq 0$, $q = \min\{m, n\}$, denote its singular values. If $Z \neq 0$, define $p_i = \sigma_i / \sum_j \sigma_j$. The effective rank is $\mathrm{EffRank}(Z) \doteq \exp\left(-\sum_{i:p_i>0} p_i \log p_i\right)$, with the convention $\mathrm{EffRank}(0) = 0$. For $\tau \in (0, 1)$, the τ-energy rank is $\mathrm{EnRank}\tau(Z) \doteq \min\left\{k : \frac{\sum_{i=1}^k \sigma_i^2}{\sum_{j=1}^r \sigma_j^2} \geq \tau\right\}$, where $r \doteq \mathrm{rank}(Z)$. Again, we assume $\mathrm{EnRank}_\tau(0) = 0$. In this paper, we use $\tau = 0.99$. Let $\phi_\theta(x) \in \mathbb{R}^d$ be the penultimate-layer representation of f_θ. For inputs $X \doteq (x_1, \ldots, x_m)$, let $\Phi_\theta(X) \in \mathbb{R}^{m\times d}$ be the representation matrix whose i-th row is $\phi_\theta(x_i)$. We consider $\mathrm{EffRank}(\Phi_\theta(X))$ and $\mathrm{EnRank}_{0.99}(\Phi_\theta(X))$ as representation-rank diagnostics.

For scalar-output networks, the empirical neural tangent kernel (eNTK) on X is $\widehat{K}(X; \theta)_{ij} \doteq \langle \nabla_\theta f_\theta(x_i), \nabla_\theta f_\theta(x_j)\rangle$. We use $\mathrm{EnRank}_{0.99}\left(\widehat{K}(X; \theta)\right)$ as the eNTK-rank diagnostic. We also compare against a curvature diagnostic, where for $H(\theta; \mathcal{T}) \doteq \nabla_\theta^2 \mathcal{L}(\theta; \mathcal{T})$, we measure $\mathrm{EnRank}_{0.99}(H(\theta; \mathcal{T}))$, or an empirical approximation thereof.

Active-neuron diagnostic. For a network with L hidden layers, let $a_i(x; \theta) \in \mathbb{R}^{w_i}$ denote the activation vector of hidden layer i. Given inputs $X \doteq \{x_1, \ldots, x_m\}$, define the relative activation score of neuron j in layer i as $s_{i,j}(X; \theta) \doteq \frac{\frac{1}{m}\sum_{x\in X} |a_i(x;\theta)_j|}{\frac{1}{m}\sum_{x\in X} \frac{1}{w_i}\sum_{r=1}^{w_i} |a_i(x;\theta)_r| + \varepsilon}$, where $\varepsilon > 0$ is a small constant to prevent division by zero. For a threshold $\tau_{\mathrm{act}} \geq 0$, the active-neuron fraction is $A_{\tau_{\mathrm{act}}}(X; \theta) \doteq \frac{\sum_{i=1}^L \sum_{j=1}^{w_i} \mathbb{I}(s_{i,j}(X;\theta) \geq \tau_{\mathrm{act}})}{\sum_{i=1}^L w_i}$, where $\mathbb{I}(\cdot)$ is the indicator function.

3 Theoretical Falsifications of Structural Plasticity Metrics

In this section, we present theoretical counterexamples to some previously adopted diagnostic metrics of plasticity. We will show, by construction, that the neural network cannot make any progress at all at a particular checkpoint in some tasks, even though those metrics would assign a favorable value to that checkpoint. In particular, the structural metrics we study in this section do not use task labels directly, and they exhibit a similar failure mode across the same checkpoint-task construction. We provide theoretical falsifications for both regression and binary classification tasks.

Shared Construction. Fix $n \geq 4$ and $M \geq n$. For binary classification, we additionally assume that n is even. Let $\bar{X} \doteq I_n$ and $\bar{x}_i \doteq e_i$ for $i = 1, \dots, n$, and consider the two-layer ReLU network $f_\theta(x) \doteq v^\top \mathrm{ReLU}(Wx)$, where $\theta \doteq (v, W)$, $v \in \mathbb{R}^M$, and $W \in \mathbb{R}^{M \times n}$. Then, for any checkpoint θ, define the representation matrix on the full support by $\Phi_\theta(\bar{X})_{i,j} \doteq \mathrm{ReLU}(w_j^\top \bar{x}_i) = \mathrm{ReLU}(W_{j,i}), \quad \Phi_\theta(\bar{X}) \in \mathbb{R}^{n \times M}$. Given a label vector $\bar{Y} \doteq [\bar{y}_1 \quad \cdots \quad \bar{y}_n]^\top$ and a differentiable pointwise loss $\ell(z, y)$, let $\bar{\mathcal{T}}(\bar{Y}, \ell)$ denote the full-support task that samples uniformly from $\{(\bar{x}_i, \bar{y}_i)\}_{i=1}^n$. Then, its population loss is $\mathcal{L}(\theta; \bar{\mathcal{T}}(\bar{Y}, \ell)) = \frac{1}{n} \sum_{i=1}^n \ell(f_\theta(\bar{x}_i), \bar{y}_i)$. We analyze the full-support gradient descent $\theta_{s+1} = \theta_s - \eta \nabla_\theta \mathcal{L}(\theta_s; \bar{\mathcal{T}}), \quad \theta_0 = \theta$, and the corresponding full-support k-step gain

$$\mathcal{G}_{\mathrm{full}}^{(k)}(\theta; \bar{\mathcal{T}}(\bar{Y}, \ell)) \doteq \begin{cases} \dfrac{\mathcal{L}(\theta; \bar{\mathcal{T}}(\bar{Y}, \ell)) - \mathcal{L}(\theta_k; \bar{\mathcal{T}}(\bar{Y}, \ell))}{\mathcal{L}(\theta; \bar{\mathcal{T}}(\bar{Y}, \ell))}, & \mathcal{L}(\theta; \bar{\mathcal{T}}(\bar{Y}, \ell)) > 0, \\ 0, & \mathcal{L}(\theta; \bar{\mathcal{T}}(\bar{Y}, \ell)) = 0. \end{cases}$$

This is a favorable setting for the learner because the update uses the exact population gradient rather than a noisy mini-batch gradient.

The following three lemmas are the only common ingredients needed for both counterexamples.

Lemma 3.1 (one step update)**.** *For a task $\bar{\mathcal{T}}(\bar{Y}, \ell)$, define $q(\theta) \doteq [q_1(\theta), \dots, q_n(\theta)]^\top$, where $q_i(\theta) \doteq \frac{\partial}{\partial z} \ell(z, \bar{y}_i)\big|_{z = f_\theta(\bar{x}_i)}$. If $\theta = (0, W)$, then $\nabla_v \mathcal{L}(\theta; \bar{\mathcal{T}}(\bar{Y}, \ell)) = \frac{1}{n} \Phi_\theta(\bar{X})^\top q(\theta)$ and $\nabla_W \mathcal{L}(\theta; \bar{\mathcal{T}}(\bar{Y}, \ell)) = 0$. Consequently, if $\Phi_\theta(\bar{X})^\top q(\theta) = 0$, then full-support gradient descent leaves θ unchanged for every learning rate $\eta > 0$, and hence $\mathcal{G}_{\mathrm{full}}^{(k)}(\theta; \bar{\mathcal{T}}(\bar{Y}, \ell)) = 0$ for all $k \geq 1$. Moreover, one full-support gradient step from $\theta_0 \doteq (0, W)$ satisfies $v_1 = -\frac{\eta}{n} \Phi_{\theta_0}(\bar{X})^\top q(\theta_0)$ and $W_1 = W$, and therefore $f_{\theta_1}(\bar{X}) = -\eta G(W) q(\theta_0)$, where $G(W) \doteq \frac{1}{n} \Phi_{\theta_0}(\bar{X}) \Phi_{\theta_0}(\bar{X})^\top$ is the Gram matrix.*

We give the proof in Appendix B.1.

Lemma 3.2 (rank metrics from a positive semi-definite activation matrix)**.** *Let $B \in \mathbb{R}^{n \times n}$ be symmetric positive semidefinite with nonnegative entries. Fix $\alpha > 0$ and suppose $v = 0$ and $\Phi_\theta(\bar{X}) = \alpha \begin{bmatrix} B & 0_{n \times (M-n)} \end{bmatrix}$. If the nonzero eigenvalues of B are $b_1 \geq b_2 \geq \cdots \geq b_r > 0$, where r is the rank of B, then the nonzero singular values of $\Phi_\theta(\bar{X})$ are $\alpha b_1, \dots, \alpha b_r$, and the nonzero singular values of the eNTK $\widehat{K}(\bar{X}; \theta)$ on $\bar{X}$ are $\alpha^2 b_1^2, \dots, \alpha^2 b_r^2$.*

The proof is in Appendix B.2.

Lemma 3.3 (random ReLU Gram concentration)**.** *Let $W_{\mathrm{rand}} \in \mathbb{R}^{M \times n}$ have i.i.d. entries sampled from*

$$\begin{cases} +\sqrt{2/n}, & \textit{with probability } 1/2, \\ -\sqrt{2/n}, & \textit{with probability } 1/2. \end{cases}$$

Let $\theta_{\mathrm{rand}} \doteq (0, W_{\mathrm{rand}})$, and define $G_{\mathrm{rand}} \doteq \frac{1}{n} \Phi_{\theta_{\mathrm{rand}}}(\bar{X}) \Phi_{\theta_{\mathrm{rand}}}(\bar{X})^\top$. Using $\mathbf{1} \in \mathbb{R}^n$ to denote the all-one vector, we have $\bar{G} \doteq \mathbb{E}[G_{\mathrm{rand}}] = \lambda(J_n + I_n)$, where $\lambda \doteq \frac{M}{2n^2}$ and $J_n = \mathbf{1}_n \mathbf{1}_n^\top$. Moreover, there is a universal constant $C > 0$, such that for every $\delta \in (0, 1)$ and every $\rho > 1$, if $M \geq C n^2 \rho \log\left(\frac{2n}{\delta}\right)$, then, with probability at least $1 - \delta$, $\left\| G_{\mathrm{rand}} - \bar{G} \right\|_2 \leq \frac{\lambda}{\sqrt{\rho}}$.

We provide the proof in Appendix B.3.

Regression Counterexample. We now construct a counterexample for the regression task, showing that there exists a checkpoint-task combination in which the model does not learn, even though the metric values are near-optimal. In addition, we present a random initialization of the hidden-layer weights that guarantees learning with high probability.

Theorem 3.4. *Let $\bar{Y}_{\mathrm{reg}} \doteq e_{n-1} - e_n$ and $\ell_{\mathrm{sq}}(z, y) \doteq \frac{1}{2}(z - y)^2$. We use $\bar{\mathcal{T}}_{\mathrm{reg}} \doteq \bar{\mathcal{T}}(\bar{Y}_{\mathrm{reg}}, \ell_{\mathrm{sq}})$ as a shorthand. There exists a checkpoint θ_{pre} such that the following hold.*

(i) The checkpoint has high representation and eNTK rank. In particular,

$$\begin{cases} \mathrm{EnRank}_{0.99}\big(\Phi_{\theta_{\mathrm{pre}}}(\bar{X})\big) & = \lceil 0.99(n-2) \rceil\,; \\ \mathrm{EffRank}\big(\Phi_{\theta_{\mathrm{pre}}}(\bar{X})\big) & = n-2; \\ \mathrm{EnRank}_{0.99}\Big(\widehat{K}(\bar{X};\theta_{\mathrm{pre}})\Big) & = \lceil 0.99(n-2) \rceil\,. \end{cases}$$

(ii) Yet, the checkpoint is completely stuck under a full-support gradient descent, where $\mathcal{L}(\theta_{\mathrm{pre}};\bar{\mathcal{T}}_{\mathrm{reg}}) = \frac{1}{n}$ and $\mathcal{G}^{(k)}_{\mathrm{full}}(\theta_{\mathrm{pre}};\bar{\mathcal{T}}_{\mathrm{reg}}) = 0$ for all $k \geq 1$ and all $\eta > 0$.

(iii) In contrast, let $\theta_{\mathrm{rand}} \doteq (0, W_{\mathrm{rand}})$ be the random initialization in Lemma 3.3. Setting $\eta = \frac{2n^2}{M}$, there exists a universal constant $C > 0$, such that for every $\delta \in (0,1)$ and $M_{\mathrm{gap}} > 1$, if $M \geq Cn^2 M_{\mathrm{gap}} \log\left(\frac{2n}{\delta}\right)$, then, with probability at least $1-\delta$, $\mathcal{G}^{(1)}_{\mathrm{full}}(\theta_{\mathrm{rand}};\bar{\mathcal{T}}_{\mathrm{reg}}) \geq 1 - \frac{1}{M_{\mathrm{gap}}}$.

The proof is in Appendix B.4.

Binary Classification Counterexample Here, we present the counterexample for the binary classification task, mirroring the goal of the regression counterexample.

Theorem 3.5. *Assume $n \geq 4$ is even. Let*

$$(\bar{Y}_{\mathrm{cls}})_i = \begin{cases} 1, & i = 1, \ldots, n/2, \\ 0, & i = n/2+1, \ldots, n, \end{cases}$$

and define the centered sign vector $\bar{S}_{\mathrm{cls}} \doteq 2\bar{Y}_{\mathrm{cls}} - \mathbf{1}_n \in \{-1,+1\}^n$. Let the binary logistic loss under the 0/1-label convention be $\ell_{\log}(z,y) \doteq \log(1+\exp(z)) - yz$, for $y \in \{0,1\}$. Equivalently, we have $\ell_{\log}(z,y) = \log\left(1+\exp\left(-(2y-1)z\right)\right)$. Define $\bar{\mathcal{T}}_{\mathrm{cls}} \doteq \bar{\mathcal{T}}(\bar{Y}_{\mathrm{cls}}, \ell_{\log})$ as a shorthand. There exists a checkpoint θ_{pre}, such that the following hold.

(i) The checkpoint has high representation and eNTK rank. In particular,

$$\begin{cases} \mathrm{EnRank}_{0.99}\big(\Phi_{\theta_{\mathrm{pre}}}(\bar{X})\big) & = \lceil 0.99(n+2) - 3 \rceil\,; \\ \mathrm{EffRank}\big(\Phi_{\theta_{\mathrm{pre}}}(\bar{X})\big) & = \frac{n}{2^{2/n}}; \\ \mathrm{EnRank}_{0.99}\Big(\widehat{K}(\bar{X};\theta_{\mathrm{pre}})\Big) & = \lceil 0.99(n+14) - 15 \rceil\,. \end{cases}$$

(ii) Yet, the checkpoint is completely stuck under a full-support gradient descent, where $\mathcal{L}(\theta_{\mathrm{pre}};\bar{\mathcal{T}}_{\mathrm{cls}}) = \log 2$ and $\mathcal{G}^{(k)}_{\mathrm{full}}(\theta_{\mathrm{pre}};\bar{\mathcal{T}}_{\mathrm{cls}}) = 0$ for all $k \geq 1$ and $\eta > 0$.

(iii) In contrast, let $\theta_{\mathrm{rand}} \doteq (0, W_{\mathrm{rand}})$ be the random initialization in Lemma 3.3. For $M_{\mathrm{gap}} > 1$, define $\gamma_{M_{\mathrm{gap}}} \doteq \log\left(\frac{2M_{\mathrm{gap}}}{\log 2}\right)$ and $\eta \doteq \frac{4\gamma_{M_{\mathrm{gap}}} n^2}{M}$. There exists a universal constant $C > 0$, such that for every $\delta \in (0,1)$, if $M \geq Cn^2\gamma^2_{M_{\mathrm{gap}}} M^2_{\mathrm{gap}} \log\left(\frac{2n}{\delta}\right)$, then, with probability at least $1-\delta$, it holds that $\mathcal{G}^{(1)}_{\mathrm{full}}(\theta_{\mathrm{rand}};\bar{\mathcal{T}}_{\mathrm{cls}}) \geq 1 - \frac{1}{M_{\mathrm{gap}}}$.

We give our proof in Appendix B.5.

Our constructions are intentionally stylized as they are not meant to model the typical trajectory of practical continual learning systems, but to isolate a failure mode showing that favorable structural diagnostic values alone do not guarantee optimization progress on a target task.

The two theorems show that high representation rank and high eNTK rank do not by themselves imply trainability on a target task. In both cases, the failure is caused by a label-dependent null direction, where the checkpoint representation has large rank on the inputs, but the task gradient is exactly orthogonal to the directions that the output layer can use at initialization. This observation motivates an optimization-derived diagnostic that depends directly on the gradient signal and its reliability, rather than only on structural properties of the checkpoint.

4 Optimization Readiness

In this section, we propose Optimization Readiness (OR) as a predictive metric of trainability. OR is the product of *gradient strength* and *gradient reliability*, where the former measures if there is a gradient signal at all, and the latter indicates the signal-to-noise ratio. Unlike diagnostics motivated primarily by structural or geometric properties of the checkpoint, OR is derived from the expected descent dynamics of stochastic gradient descent. This connection allows us to prove a lower bound on one-step gain under a smoothness assumption and an appropriate learning rate.

Let $\mathcal{T} \doteq (\mathcal{X}, \mathcal{Y}, P_{\mathcal{XY}})$ be an arbitrary supervised learning task and $\mathcal{B}$ be a mini-batch of size m, where each input-label pair from $\mathcal{B}$ is drawn i.i.d. from $P_{\mathcal{XY}}$. We formally state the assumptions and define OR as follows.

Assumption 4.1 (unbiasedness)**.** For any checkpoint θ, it holds that $\mathbb{E}_{\mathcal{B}}[\widehat{g}_{\mathcal{B}}(\theta; \mathcal{T})] = g(\theta; \mathcal{T})$.

Assumption 4.2 (smoothness)**.** The population loss $\mathcal{L}(\theta; \mathcal{T})$ is differentiable with respect to θ and β-smooth, such that for all θ_1, θ_2, it holds that $\|\nabla_\theta \mathcal{L}(\theta_1; \mathcal{T}) - \nabla_\theta \mathcal{L}(\theta_2; \mathcal{T})\|_2 \leq \beta \|\theta_1 - \theta_2\|_2$ for some $\beta > 0$.

The Euclidean norm of the gradient vector is a straightforward way to monitor the magnitude of the gradient signal. When the gradient norm becomes zero, gradient descent stops updating the parameters. In addition, gradient norm alone can be misleading because it does not distinguish checkpoints at different stages of convergence. Hence, we normalize the squared gradient norm by the population loss and define gradient strength as

Definition 4.3 (gradient strength)**.** $S(\theta; \mathcal{T}) \doteq \frac{\|g(\theta;\mathcal{T})\|_2^2}{\mathcal{L}(\theta;\mathcal{T})}$.

The noise in gradient estimation can also dictate the progress of gradient descent, as discussed in Baveja et al. [2025]. Gradient descent may fail to reduce population loss if the gradient signal is overwhelmed by noise, even when the gradient is strong. In light of this observation, we define gradient reliability as

Definition 4.4 (gradient reliability)**.** $R(\theta; \mathcal{T}) \doteq \frac{\|g(\theta;\mathcal{T})\|_2^2}{\|g(\theta;\mathcal{T})\|_2^2 + V_{\mathcal{B}}(\theta;\mathcal{T})}$, where $V_{\mathcal{B}}(\theta; \mathcal{T}) \doteq \mathbb{E}_{\mathcal{B}}\left[\|\widehat{g}_{\mathcal{B}}(\theta; \mathcal{T}) - g(\theta; \mathcal{T})\|_2^2\right]$.

Notably, if Assumption 4.1 holds, then $R(\theta; \mathcal{T}) = \frac{\|g(\theta;\mathcal{T})\|_2^2}{\|g(\theta;\mathcal{T})\|_2^2 + \left(\mathbb{E}_{\mathcal{B}}\left[\|\widehat{g}_{\mathcal{B}}(\theta;\mathcal{T})\|_2^2\right] - \|g(\theta;\mathcal{T})\|_2^2\right)} = \frac{\|g(\theta;\mathcal{T})\|_2^2}{\mathbb{E}_{\mathcal{B}}\left[\|\widehat{g}_{\mathcal{B}}(\theta;\mathcal{T})\|_2^2\right]}$. We assume the mini-batch size to estimate $R(\theta; \mathcal{T})$ is consistent with the gradient descent procedure. We therefore define

Definition 4.5 (Optimization Readiness)**.**

$$\mathrm{OR}(\theta; \mathcal{T}) \doteq \begin{cases} S(\theta; \mathcal{T}) R(\theta; \mathcal{T}), & \mathcal{L}(\theta; \mathcal{T}) > 0 \text{ and } \mathbb{E}_{\mathcal{B}}\left[\|\widehat{g}_{\mathcal{B}}(\theta; \mathcal{T})\|_2^2\right] > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Theorem 4.6 (Optimization Readiness Lower-Bounds One-Step Gain)**.** *Let $\mathcal{T} \doteq (\mathcal{X}, \mathcal{Y}, P_{\mathcal{XY}})$ be any supervised learning task and θ be an arbitrary parameterization of the neural network. Let $\theta_0 \doteq \theta$ and θ_1 be the parameters after one step of mini-batch gradient descent as in (1) with $\mathcal{B}_0 \doteq \{(x_1, y_1), \ldots, (x_m, y_m)\}$, where $(x_i, y_i) \overset{i.i.d.}{\sim} P_{\mathcal{XY}}$ for $i = 1, \ldots, m$. Suppose Assumption 4.1 and 4.2 hold. If $0 < \eta < R(\theta; \mathcal{T})/\beta$, then it holds that $\mathcal{G}^{(1)}(\theta; \mathcal{T}) \geq \frac{\alpha(1-\alpha)}{\beta} \mathrm{OR}(\theta; \mathcal{T})$, where $\alpha = \eta\beta / R(\theta; \mathcal{T})$ and β is the smoothness constant.*

The proof can be found in Appendix B.6.

5 Experiments

In this section, we empirically verify if the metrics of plasticity can reliably predict k-step gains. We use Slowly-Changing Regression (SCR) for continual regression and Permuted MNIST (P-MNIST) for continual classification as testbeds [Dohare et al., 2024]. For SCR, the features are binary vectors of dimension $u + v$, where the first u dimensions are slowly-changing bits, and the remaining v dimensions are random bits. Suppose each task has N data points, the v random bits are randomly sampled for each data point, while the u slowly changing bits stay constant for N steps. After N steps, one of the u bits is uniformly randomly selected and flipped from 1 to 0 or from 0 to 1, thereby

Table 1: Dataset generation statistics for Slowly-Changing Regression and Permuted MNIST.

	Slowly-Changing Regression	Permuted MNIST
Training tasks per sequence	1,000	800
Training samples per sequence	1,000,000	8,000,000
Validation samples per task	10,000	10,000
Training sequences	20	20
Validation tasks	30	10
Task shift mechanism	Bit flipping every 1,000 samples	Random pixel permutation per task

creating a slowly shifting distribution in the input space across tasks. The labels are generated by a linear threshold unit (LTU) network [Dohare et al., 2024] with its weights randomly initialized to be $+1$ or -1 and fixed across all tasks. Regarding P-MNIST, we generate a random permutation of the pixels for each task and apply it to all images within that task, while keeping the original labels. We flatten the images row-wise to form the feature vectors.

We compare OR against representative structural diagnostics, including representation-rank metrics, eNTK rank, and active-neuron fraction. We also include Hessian rank as a task-dependent curvature baseline, allowing us to test whether OR remains useful beyond purely structural input-checkpoint diagnostics.

Data Generation. We generate separate training and validation tasks for both SCR and P-MNIST. The training tasks are designed to induce loss of plasticity through continual adaptation to non-stationary data streams, while the validation tasks are used to evaluate the predictive power of plasticity metrics on held-out stationary tasks. Detailed dataset statistics are summarized in Table 1.

Checkpoint Generation. We initialize and train a multi-layer perceptron (MLP) with ReLU activation on each training sequence of SCR and P-MNIST. The purpose of training is to obtain a set of checkpoints with varying trainabilities for later testing of the metrics. The MLP for SCR has 2 hidden layers of 5 units, whereas the MLP for P-MNIST has 3 hidden layers of 100 units. We employ mean-squared error for SCR and cross-entropy loss for P-MNIST without regularization. We use Adam [Kingma and Ba, 2015] as the optimizer with batch size $m = 1$. The learning rates are set to $\eta = 0.01$ and $\eta = 0.003$ for SCR and P-MNIST, respectively. Figure 1 shows the learning curves for both tasks. There is a clear sign of plasticity loss in both scenarios. As we save a checkpoint every 5 tasks, we collect a wide range of checkpoints with varying levels of trainability.

k-Step Gain and Metric Estimation. Given a validation task $\mathcal{T}_{\text{val}}$ and a checkpoint θ, we estimate the k-step gain by running stochastic gradient descent for k steps from θ on mini-batches sampled from the validation set, and then measuring the resulting relative reduction in empirical loss. We repeat this procedure over multiple independent optimization rollouts to approximate the expectation in Definition 2.1.

We estimate OR using empirical counterparts of its two components. The population gradient is approximated by the gradient of the full validation set, while the expected mini-batch squared gradient norm is estimated using independently sampled mini-batches from the validation set. Together, these quantities give empirical estimates of gradient strength, gradient reliability, and OR.

For the baseline diagnostics, we compute representation-rank metrics, active-neuron fraction, eNTK rank, and Hessian-rank metrics on the same validation data whenever applicable. For SCR, the network is small enough to compute the exact Hessian and eNTK rank. For P-MNIST, we omit the eNTK rank because the vector-valued output makes the corresponding empirical kernel less directly comparable to the scalar-output case, and we approximate the Hessian rank using the Gram-matrix estimator of Lewandowski et al. [2024]. Full estimator definitions and implementation details are provided in Appendix C.1.

Qualitative Analysis. We first conduct a qualitative analysis to examine the relationship between the metric values and their corresponding k-step gains. Figure 2 displays the metric values against the ground-truth k-step gains. We observe an approximately linear relationship between the 1-step gain and OR, whereas the other metrics do not exhibit a clear positive correlation with trainability. Even for $k = 100$, OR maintains a strong positive relationship with the k-step gain, suggesting that it remains informative beyond the one-step setting considered in our theory.

Table 2: Pairwise ranking accuracy (mean $\pm$ standard error) across tasks.

Task	Metric	$k=1$	$k=10$	$k=100$
Slowly-Changing Regression	OR	**0.987 $\pm$ 0.001**	**0.988 $\pm$ 0.001**	**0.977 $\pm$ 0.002**
	Repr Effective Rank	0.698 $\pm$ 0.001	0.700 $\pm$ 0.001	0.704 $\pm$ 0.001
	Repr 99% Energy Rank	0.689 $\pm$ 0.002	0.690 $\pm$ 0.002	0.694 $\pm$ 0.002
	eNTK 99% Energy Rank	0.765 $\pm$ 0.002	0.768 $\pm$ 0.002	0.779 $\pm$ 0.003
	Hessian 99% Energy Rank	0.734 $\pm$ 0.003	0.737 $\pm$ 0.003	0.746 $\pm$ 0.005
	Active Neuron Fraction	0.671 $\pm$ 0.007	0.673 $\pm$ 0.007	0.679 $\pm$ 0.007
Permuted MNIST	OR	**0.906 $\pm$ 0.005**	**0.905 $\pm$ 0.005**	**0.899 $\pm$ 0.004**
	Repr Effective Rank	0.858 $\pm$ 0.002	0.864 $\pm$ 0.003	0.873 $\pm$ 0.002
	Repr 99% Energy Rank	0.828 $\pm$ 0.003	0.834 $\pm$ 0.003	0.844 $\pm$ 0.003
	Hessian 99% Energy Rank	0.821 $\pm$ 0.003	0.753 $\pm$ 0.007	0.717 $\pm$ 0.007
	Active Neuron Fraction	0.346 $\pm$ 0.003	0.377 $\pm$ 0.003	0.391 $\pm$ 0.003

Next, we visualize the evolution of the k-step gains and the metric values throughout continual training in Figure 3, where the top three rows correspond to the k-step gains and the bottom row shows the metric trajectories. Among all metrics, the trajectory of OR aligns most closely with the evolution of the k-step gains, particularly in SCR. Interestingly, trainability is not maximized near random initialization. Instead, the k-step gains initially increase rapidly before gradually declining. We conjecture that this phenomenon may be related to forward transfer [Lopez-Paz and Ranzato, 2017], where previously learned tasks temporarily facilitate adaptation to similar future tasks before the effects of plasticity loss begin to dominate.

Quantitative Analysis. We next quantitatively evaluate whether each metric preserves the ordering of checkpoints by trainability. Let $\omega(\theta;\mathcal{T})$ denote the value of a diagnostic metric at checkpoint θ on task $\mathcal{T}$. For two checkpoints θ_1 and θ_2, a metric ranks them correctly if its ordering agrees with the ordering induced by the k-step gain, i.e., $(\omega(\theta_1;\mathcal{T}) - \omega(\theta_2;\mathcal{T}))\big(\mathcal{G}^{(k)}(\theta_1;\mathcal{T}) - \mathcal{G}^{(k)}(\theta_2;\mathcal{T})\big) > 0$. We therefore measure pairwise ranking accuracy, defined as the fraction of checkpoint pairs whose metric ordering agrees with their ordering under the estimated ground-truth k-step gain:

$$\frac{\sum_{(\theta_1,\theta_2)\in\tilde{\Theta}\times\tilde{\Theta}} \mathbb{I}(\Delta_\omega(\theta_1,\theta_2;\mathcal{T})\Delta_{\mathcal{G}^{(k)}}(\theta_1,\theta_2;\mathcal{T}) > 0)\mathbb{I}(\Delta_{\mathcal{G}^{(k)}}(\theta_1,\theta_2;\mathcal{T}) \neq 0)}{\sum_{(\theta_1,\theta_2)\in\tilde{\Theta}\times\tilde{\Theta}} \mathbb{I}(\Delta_{\mathcal{G}^{(k)}}(\theta_1,\theta_2;\mathcal{T}) \neq 0)},$$

where $\tilde{\Theta}$ is the set of recorded checkpoints, $\Delta_\omega(\theta_1,\theta_2;\mathcal{T}) \doteq \omega(\theta_1;\mathcal{T}) - \omega(\theta_2;\mathcal{T})$, and $\Delta_{\mathcal{G}^{(k)}}(\theta_1,\theta_2;\mathcal{T}) \doteq \mathcal{G}^{(k)}(\theta_1;\mathcal{T}) - \mathcal{G}^{(k)}(\theta_2;\mathcal{T})$. This quantity is closely related to Kendall-style rank correlation measures [Kendall, 1938], but has a direct probabilistic interpretation as the probability that a metric correctly ranks two checkpoints by trainability.

We collect 201 checkpoints per SCR run and 161 per P-MNIST run. Table 2 reports the pairwise ranking accuracies across tasks and horizons. OR achieves the highest ranking accuracy for both tasks and all values of k, with the largest margin appearing in SCR.

Subsampling Ablation. The preceding results estimate all metrics using the full validation set. However, access to large held-out datasets may be unrealistic in practical continual learning scenarios. We therefore evaluate whether OR remains predictive when estimated from substantially fewer samples. For SCR, we estimate OR using 10%, 1%, and 0.1% of the validation set, corresponding to 1,000, 100, and 10 data points, respectively. For P-MNIST, we use 10% and 1% of the validation set, corresponding to 1,000 and 100 data points. In all subsampling experiments, we estimate OR using only 16 mini-batches of size $m = 1$.

Figure 4 summarizes the results. In SCR, OR remains the most accurate metric even when estimated from only 0.1% of the validation data, suggesting that the full-data configuration used above is conservative. In P-MNIST, the accuracy of OR decreases under subsampling, especially for larger k, and can fall below representation-rank metrics computed using the full validation set. Nevertheless, it remains the strongest predictor among the metrics compared under the same data budget. This observation also suggests that OR can remain informative even when estimated using mini-batches smaller than those used in the k-step gain rollouts.

6 Related Work

Explanations and Diagnostics of Loss of Plasticity. Understanding and diagnosing loss of plasticity has been an active area of research in both continual supervised and reinforcement learning settings [Dohare et al., 2024, Lyle et al., 2025]. A prominent line of work studies the structural properties of neural representations. In particular, representation collapse has been proposed as a key mechanism underlying plasticity loss, where metrics such as the effective rank and the 99% energy rank of the representation are used to explain and detect degraded trainability [Dohare et al., 2023, 2024]. Another widely studied class of diagnostics focuses on spectral properties of the neural tangent kernel (NTK), including its rank, sharpness, and condition number [Lyle et al., 2025, Tang et al., 2025], where ill-conditioned NTKs are associated with optimization difficulty. Loss of curvature directions, reflected by a collapse of the Hessian rank, has also been proposed as an explanation for plasticity loss [Lewandowski et al., 2024, He et al., 2025]. More recently, Wang et al. [2026] study finite-time Lyapunov exponents (FTLEs), arguing that both excessively small and excessively large FTLEs can impair plasticity. On a more microscopic level, dormant or dead neurons are also used as indicators of plasticity loss [Sokar et al., 2023, Ma et al., 2024].

While these works provide valuable explanations and empirical correlations, many existing diagnostics are motivated primarily through structural or geometric properties of the checkpoint, rather than through a direct characterization of future optimization progress. It therefore remains unclear whether such diagnostics can reliably predict future trainability. Our work studies this predictive question directly by evaluating diagnostics against future optimization gain and by proposing an optimization-derived metric with theoretical and empirical support.

Empirical Falsifications and Limitations of Existing Diagnostics. Several works have demonstrated that existing diagnostics cannot fully explain trainability in neural networks. Lyle et al. [2023] show that metrics such as weight norm, weight rank, dead units, and feature rank can exhibit inconsistent correlations with plasticity across settings. Lewandowski et al. [2024] construct empirical counterexamples showing that update norm, neuron dormancy, representation rank, and weight norm can all be misleading indicators of plasticity. Likewise, Baveja et al. [2025] demonstrate that individual indicators such as Hessian rank, unit sign entropy, and gradient norm cannot fully characterize trainability on their own.

These works suggest that existing diagnostics are incomplete, but their analysis is primarily empirical. Our work complements this line of research by providing theoretical counterexamples that explicitly characterize failure modes of several widely used diagnostics.

7 Discussion and Conclusion

In this work, we study plasticity in deep continual learning from the perspective of future optimization progress. We propose k-step gain as an operational proxy for trainability and use it to evaluate whether existing diagnostics can predict how much a checkpoint will improve after further optimization on a target task.

Our theoretical results show that high structural diversity alone does not guarantee trainability. In particular, we construct regression and binary classification tasks on which metrics based on representation rank and eNTK rank assign favorable values to a checkpoint, even though full-support gradient descent cannot reduce the loss from that checkpoint. On the same tasks, we also show that randomly initialized weights are likely to make substantial one-step progress, highlighting a concrete gap between structural diagnostic values and actual optimization gain.

Motivated by this gap, we propose optimization readiness, an optimization-derived metric combining gradient strength and gradient reliability. Under a smoothness assumption and an appropriate learning rate, we prove that the optimization readiness lower bounds the one-step gain. Empirically, we find that OR reliably ranks checkpoints by their 1-, 10-, and 100-step gains on both Slowly-Changing Regression and Permuted MNIST, outperforming several commonly used diagnostics. Our subsampling ablation further suggests that OR can remain informative even when estimated with limited validation data and a small number of Monte Carlo samples.

Our study deliberately focuses on simple yet standard continual learning testbeds in order to isolate the predictive behavior of plasticity diagnostics. This controlled setting allows us to connect theory and empirical evaluation cleanly, but it also leaves several important directions for future work. First, our counterexamples use ReLU networks with one hidden layer. Extending the analysis to deeper networks and other activation functions would strengthen the theoretical scope. Second, our empirical evaluation focuses on supervised continual learning. Thus, testing whether the same observations hold in continual reinforcement learning and larger-scale architectures and problems is an important next step. Overall, our results suggest that diagnostics of plasticity should be evaluated not only as retrospective explanations of observed training behavior, but also as prospective predictors of future optimization gain.

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A Math Background

Theorem A.1. *(Matrix Bernstein inequality) Let $M_1, \ldots, M_k$ be independent random symmetric matrices such that $M_i \in \mathbb{R}^{n\times n}$, $\mathbb{E}[M_i] = 0$, and $\|M_i\| \leq C$ almost surely for all $i \in \{1, \ldots, k\}$. Then, for all $t \geq 0$, it holds that*

$$\Pr\left(\left\|\sum_{i=1}^{k} M_i\right\| \geq t\right) \leq 2n \exp\left(\frac{-t^2}{2(\sigma^2 + Ct/3)}\right),$$

where $\sigma^2 \doteq \left\|\sum_{i=1}^{k} \mathbb{E}[M_i^2]\right\|$.

B Proof of Theoretical Results

B.1 Proof of Lemma 3.1

Proof. For any x, we have

$$\nabla_v f_\theta(x) = \mathrm{ReLU}(Wx)$$

and

$$\nabla_{w_j} f_\theta(x) = v_j \mathbb{I}\big(w_j^\top x > 0\big)x.$$

Therefore,

$$\nabla_v \mathcal{L}(\theta; \bar{\mathcal{T}}(\bar{Y}, \ell)) = \frac{1}{n}\sum_{i=1}^{n} q_i(\theta)\mathrm{ReLU}(W\bar{x}_i) = \frac{1}{n}\Phi_\theta(\bar{X})^\top q(\theta),$$

and, for each hidden unit j,

$$\nabla_{w_j} \mathcal{L}(\theta; \bar{\mathcal{T}}(\bar{Y}, \ell)) = \frac{1}{n}\sum_{i=1}^{n} q_i(\theta) v_j \mathbb{I}\big(w_j^\top \bar{x}_i > 0\big)\bar{x}_i.$$

If $v = 0$, then every first-layer gradient is zero. If additionally $\Phi_\theta(\bar{X})^\top q(\theta) = 0$, then the output-layer gradient is also zero, so $\nabla_\theta \mathcal{L}(\theta; \bar{\mathcal{T}}(\bar{\mathcal{Y}}, \ell)) = 0$. Hence gradient descent leaves θ unchanged for all time, which implies zero full-support gain.

For the one-step update formula, when $\theta_0 = (0, W)$, the first-layer gradient is zero and

$$\begin{cases} v_1 & = -\frac{\eta}{n}\Phi_{\theta_0}(\bar{X})^\top q(\theta_0), \\ W_1 & = W. \end{cases}$$

Thus, we have

$$f_{\theta_1}(\bar{X}) = \Phi_{\theta_0}(\bar{X})v_1 = -\frac{\eta}{n}\Phi_{\theta_0}(\bar{X})\Phi_{\theta_0}(\bar{X})^\top q(\theta_0) = -\eta G(W)q(\theta_0).$$

□

B.2 Proof of Lemma 3.2

Proof. Since B is symmetric positive semidefinite, its singular values are exactly its eigenvalues. Appending zero columns and multiplying by α gives nonzero singular values

$$\alpha b_1, \dots, \alpha b_r$$

for $\Phi_\theta(\bar{X})$. Since $v = 0$, the first-layer portion of $\nabla_\theta f_\theta(\bar{x}_i)$ is zero for every i. Hence, the eNTK is generated only by output-layer gradients, and we have

$$\widehat{K}(\bar{X}; \theta) = \Phi_\theta(\bar{X})\Phi_\theta(\bar{X})^\top = \alpha^2 B^2.$$

The nonzero eigenvalues, and thus the nonzero singular values, of this positive semidefinite matrix are

$$\alpha^2 b_1^2, \dots, \alpha^2 b_r^2.$$

□

B.3 Proof of Lemma 3.3

Proof. Define $a \doteq \sqrt{2/n}$ for simplicity. For each hidden unit j, define $h_j \in \mathbb{R}^n$ by

$$(h_j)_i \doteq \mathrm{ReLU}((W_{\mathrm{rand}})_{j,i}).$$

Then $(h_j)_i = a$ with probability $1/2$ and 0 with probability $1/2$, independently over i. Therefore, it holds that

$$G_{\mathrm{rand}} = \frac{1}{n}\sum_{j=1}^{M} h_j h_j^\top.$$

For $i \neq k$, we have

$$\mathbb{E}[(h_j)_i(h_j)_k] = \mathbb{E}[(h_j)_i]\mathbb{E}[(h_j)_k] = \frac{a^2}{4} = \frac{1}{2n},$$

while

$$\mathbb{E}\big[(h_j)_i^2\big] = \frac{a^2}{2} = \frac{1}{n}.$$

Thus, we have

$$\mathbb{E}\big[h_j h_j^\top\big] = \frac{1}{2n}(J_n + I_n)$$

and

$$\bar{G} \doteq \mathbb{E}[G_{\mathrm{rand}}] = \frac{M}{2n^2}(J_n + I_n) = \lambda(J_n + I_n).$$

It remains to prove concentration. Let $A_j \doteq \frac{1}{n}h_j h_j^\top$ and $Q_j \doteq A_j - \mathbb{E}[A_j]$. Then, $G_{\mathrm{rand}} - \bar{G} = \sum_{j=1}^{M} Q_j$, and the Q_j's are independent, zero-mean, symmetric matrices. Since $\|h_j\|_2^2 \leq na^2 = 2$, it holds that

$$\|A_j\|_2 = \frac{1}{n}\|h_j\|_2^2 \leq \frac{2}{n}.$$

By Jensen's inequality, $\|\mathbb{E}[A_j]\|_2 \leq 2/n$, and hence $\|Q_j\|_2 \leq \frac{4}{n}$. Moreover, we have

$$\left\|\mathbb{E}[Q_j^2]\right\|_2 \leq \left\|\mathbb{E}[A_j^2]\right\|_2 \leq \mathbb{E}[\|A_j\|_2^2] \leq \frac{4}{n^2}.$$

Therefore, the matrix-variance parameter satisfies

$$\left\|\sum_{j=1}^{M} \mathbb{E}[Q_j^2]\right\|_2 \leq \frac{4M}{n^2}.$$

By Matrix Bernstein (Theorem A.1), there is a universal constant $C_0 > 0$, such that with probability at least $1-\delta$,

$$\left\|G_{\text{rand}} - \bar{G}\right\|_2 \leq C_0 \left(\frac{\sqrt{M \log(2n/\delta)}}{n} + \frac{\log(2n/\delta)}{n} \right).$$

Let $z \doteq \log(2n/\delta)$. Since $\lambda = M/(2n^2)$, the last display gives

$$\frac{\left\|G_{\text{rand}} - \bar{G}\right\|_2}{\lambda} \leq 2C_0 \frac{n\sqrt{z}}{\sqrt{M}} + 2C_0 \frac{nz}{M}.$$

If $M \geq Cn^2\rho z$, then, after increasing the universal constant C if necessary, the right-hand side is at most $1/\sqrt{\rho}$. Consequently, we have

$$\left\|G_{\text{rand}} - \bar{G}\right\|_2 \leq \frac{\lambda}{\sqrt{\rho}},$$

as claimed. □

B.4 Proof of Theorem 3.4

Proof. Let

$$B_{\text{reg}} \doteq \begin{bmatrix} I_{n-2} & 0_{(n-2)\times 2} \\ 0_{2\times(n-2)} & 0_{2\times 2} \end{bmatrix}$$

and $\alpha \doteq \frac{1}{2}$. Choose $v_{\text{pre}} = 0$ and choose W_{pre}, such that

$$\Phi_{\theta_{\text{pre}}}(\bar{X}) = \alpha \begin{bmatrix} B_{\text{reg}} & 0_{n\times(M-n)} \end{bmatrix}.$$

This is realized, for example, by setting

$$(W_{\text{pre}})_{j,i} = \begin{cases} \alpha, & j = i \leq n-2, \\ -1, & \text{otherwise.} \end{cases}$$

The nonzero eigenvalues of B_{reg} are all equal to 1, with multiplicity $n-2$. Lemma 3.2 therefore implies that the nonzero singular values of $\Phi_{\theta_{\text{pre}}}(\bar{X})$ are

$$\underbrace{\alpha, \ldots, \alpha}_{n-2 \text{ times}},$$

and the nonzero singular values of $\widehat{K}(\bar{X}; \theta_{\text{pre}})$ are

$$\underbrace{\alpha^2, \ldots, \alpha^2}_{n-2 \text{ times}}.$$

Thus, we have

$$\begin{cases} \text{EnRank}_{0.99}\big(\Phi_{\theta_{\text{pre}}}(\bar{X})\big) & = \lceil 0.99(n-2) \rceil; \\ \text{EnRank}_{0.99}\Big(\widehat{K}(\bar{X}; \theta_{\text{pre}})\Big) & = \lceil 0.99(n-2) \rceil. \end{cases}$$

The normalized nonzero singular values of $\Phi_{\theta_{\text{pre}}}(\bar{X})$ are all $1/(n-2)$, so

$$\text{EffRank}\big(\Phi_{\theta_{\text{pre}}}(\bar{X})\big) = \exp\left(-\sum_{i=1}^{n-2} \frac{1}{n-2} \log \frac{1}{n-2} \right) = n-2.$$

This proves part (i) of Theorem 3.4.

For the squared loss, at a zero-output checkpoint, we have

$$q(\theta_{\text{pre}}) = -\bar{Y}_{\text{reg}}.$$

Since $B_{\text{reg}}\bar{Y}_{\text{reg}} = 0$, we have

$$\Phi_{\theta_{\text{pre}}}(\bar{X})^\top q(\theta_{\text{pre}}) = -\alpha \begin{bmatrix} B_{\text{reg}} \\ 0 \end{bmatrix} \bar{Y}_{\text{reg}} = 0.$$

Lemma 3.1 implies that full-support gradient descent leaves θ_{pre} unchanged for every learning rate. In addition, it holds that

$$\mathcal{L}(\theta_{\text{pre}}; \bar{\mathcal{T}}_{\text{reg}}) = \frac{1}{2n}\left\|\bar{Y}_{\text{reg}}\right\|_2^2 = \frac{1}{n}.$$

Hence, we have $\mathcal{G}^{(k)}_{\text{full}}(\theta_{\text{pre}}; \bar{\mathcal{T}}_{\text{reg}}) = 0$ for all $k \geq 1$, proving part (ii) of Theorem 3.4.

It remains to prove the random-initialization claim. Let $\theta_0 = \theta_{\text{rand}}$ and G_{rand} be as in Lemma 3.3, and set $\lambda = \frac{M}{2n^2}$ and $\eta = \frac{1}{\lambda} = \frac{2n^2}{M}$. By Lemma 3.1, since $q(\theta_0) = -\bar{Y}_{\text{reg}}$, it holds that

$$f_{\theta_1}(\bar{X}) = \eta G_{\text{rand}} \bar{Y}_{\text{reg}}.$$

Furthermore, since $\bar{Y}_{\text{reg}} \perp \mathbf{1}_n$, we get

$$\bar{G}\bar{Y}_{\text{reg}} = \lambda \bar{Y}_{\text{reg}}.$$

Therefore, we have

$$f_{\theta_1}(\bar{X}) - \bar{Y}_{\text{reg}} = \eta(G_{\text{rand}} - \bar{G})\bar{Y}_{\text{reg}}.$$

Applying Lemma 3.3 with $\rho = M_{\text{gap}}$, the stated width condition implies that, with probability at least $1 - \delta$,

$$\left\|G_{\text{rand}} - \bar{G}\right\|_2 \leq \frac{\lambda}{\sqrt{M_{\text{gap}}}}.$$

On this event, it holds that

$$\mathcal{L}\big(\theta_1; \bar{\mathcal{T}}_{\text{reg}}\big) = \frac{1}{2n}\left\|f_{\theta_1}(\bar{X}) - \bar{Y}_{\text{reg}}\right\|_2^2 \leq \frac{1}{2n}\frac{1}{M_{\text{gap}}}\left\|\bar{Y}_{\text{reg}}\right\|_2^2 = \frac{1}{M_{\text{gap}}}\mathcal{L}\big(\theta_0; \bar{\mathcal{T}}_{\text{reg}}\big).$$

As a result, we have

$$\mathcal{G}^{(1)}_{\text{full}}(\theta_{\text{rand}}; \bar{\mathcal{T}}_{\text{reg}}) = \frac{\mathcal{L}(\theta_0; \bar{\mathcal{T}}_{\text{reg}}) - \mathcal{L}(\theta_1; \bar{\mathcal{T}}_{\text{reg}})}{\mathcal{L}(\theta_0; \bar{\mathcal{T}}_{\text{reg}})} \geq 1 - \frac{1}{M_{\text{gap}}}.$$

$\square$

B.5 Proof of Theorem 3.5

Proof. For simplicity, we write $\bar{Y} = \bar{Y}_{\text{cls}}$ and $\bar{S} = \bar{S}_{\text{cls}} = 2\bar{Y} - \mathbf{1}_n$. By construction, it holds that $\bar{S}^\top \mathbf{1}_n = 0$ and $\left\|\bar{S}\right\|_2^2 = n$. Define $B_{\text{cls}} \doteq I_n + \frac{1}{n}\big(\mathbf{1}_n\mathbf{1}_n^\top - \bar{S}\bar{S}^\top\big)$ and $\alpha \doteq \frac{1}{2}$. Equivalently, we have

$$(B_{\text{cls}})_{i,j} = \begin{cases} 1, & i = j, \\ 0, & i \neq j \text{ and } \bar{y}_i = \bar{y}_j, \\ 2/n, & \bar{y}_i \neq \bar{y}_j. \end{cases}$$

Thus, B_{cls} has nonnegative entries. Choose $v_{\text{pre}} = 0$ and choose W_{pre}, such that

$$\Phi_{\theta_{\text{pre}}}(\bar{X}) = \alpha \begin{bmatrix} B_{\text{cls}} & 0_{n\times(M-n)} \end{bmatrix}.$$

One concrete realization is to set, for $j \leq n$,

$$(W_{\text{pre}})_{j,i} = \begin{cases} \alpha(B_{\text{cls}})_{i,j}, & (B_{\text{cls}})_{i,j} > 0, \\ -1, & (B_{\text{cls}})_{i,j} = 0, \end{cases}$$

and to set $(W_{\text{pre}})_{j,i} = -1$ for all $j > n$.

The eigenspaces of B_{cls} are

$$\begin{cases} B_{\text{cls}}\bar{S} &= 0; \\ B_{\text{cls}}\mathbf{1}_n &= 2\mathbf{1}_n; \\ B_{\text{cls}}u &= u \end{cases}$$

for every $u \perp \operatorname{span}\{\mathbf{1}_n, \bar{S}\}$. Hence, the eigenvalues of B_{cls} are

$$2, \quad \underbrace{1, \ldots, 1}_{n-2 \text{ times}}, \quad 0.$$

By Lemma 3.2, the nonzero singular values of $\Phi_{\theta_{\text{pre}}}(\bar{X})$ are

$$2\alpha, \quad \underbrace{\alpha, \ldots, \alpha}_{n-2 \text{ times}}.$$

Therefore, for $1 \le k \le n-1$, the fraction of squared singular-value energy captured by the top k singular values is

$$\frac{(2\alpha)^2 + (k-1)\alpha^2}{(2\alpha)^2 + (n-2)\alpha^2} = \frac{k+3}{n+2}.$$

The smallest k for which this is at least 0.99 is $\lceil 0.99(n+2) - 3 \rceil$. For the effective rank, the normalized, nonzero singular values are

$$\frac{2}{n}, \quad \underbrace{\frac{1}{n}, \ldots, \frac{1}{n}}_{n-2 \text{ times}}.$$

Therefore, we have

$$\operatorname{EffRank}\big(\Phi_{\theta_{\text{pre}}}(\bar{X})\big) = \exp\left(-\frac{2}{n}\log\frac{2}{n} - (n-2)\frac{1}{n}\log\frac{1}{n}\right) = \frac{n}{2^{2/n}}.$$

Similarly, the nonzero singular values of $\widehat{K}(\bar{X}; \theta_{\text{pre}})$ are

$$4\alpha^2, \quad \underbrace{\alpha^2, \ldots, \alpha^2}_{n-2 \text{ times}}.$$

Thus, for $1 \le k \le n-1$, the fraction of squared singular-value energy captured by the top k eNTK singular values is

$$\frac{(4\alpha^2)^2 + (k-1)(\alpha^2)^2}{(4\alpha^2)^2 + (n-2)(\alpha^2)^2} = \frac{k+15}{n+14},$$

and hence

$$\operatorname{EnRank}_{0.99}\Big(\widehat{K}(\bar{X}; \theta_{\text{pre}})\Big) = \lceil 0.99(n+14) - 15 \rceil.$$

This proves part (i) of Theorem 3.5.

For the $0/1$-label logistic loss,

$$\frac{\partial}{\partial z}\ell_{\log}(z, y) = \frac{\exp(z)}{1+\exp(z)} - y.$$

At a zero-output checkpoint, this gives $q_i(\theta_{\text{pre}}) = \frac{1}{2} - \bar{y}_i = -\frac{1}{2}\bar{s}_i$, and $q(\theta_{\text{pre}}) = -\frac{1}{2}\bar{S}$. Since $B_{\text{cls}}\bar{S} = 0$, then we get

$$\Phi_{\theta_{\text{pre}}}(\bar{X})^\top q(\theta_{\text{pre}}) = -\frac{\alpha}{2}\begin{bmatrix} B_{\text{cls}} \\ 0 \end{bmatrix}\bar{S} = 0.$$

Lemma 3.1 implies that full-support gradient descent leaves θ_{pre} unchanged for every learning rate. Since $f_{\theta_{\text{pre}}}(\bar{x}_i) = 0$ for all i, we get

$$\mathcal{L}(\theta_{\text{pre}}; \bar{\mathcal{T}}_{\text{cls}}) = \frac{1}{n}\sum_{i=1}^{n}\log 2 = \log 2.$$

Consequently, we have

$$\mathcal{G}^{(k)}_{\text{full}}(\theta_{\text{pre}}; \bar{\mathcal{T}}_{\text{cls}}) = 0$$

for all $k \geq 1$ proving part (ii) of Theorem 3.5.

It remains to prove the random-initialization claim. Let $\theta_0 = \theta_{\text{rand}}$ and G_{rand} be as in Lemma 3.3. Set $\lambda = \frac{M}{2n^2}$. At θ_0, the prediction is zero and $q(\theta_0) = -\frac{1}{2}\bar{S}$. By Lemma 3.1, we have

$$f_{\theta_1}(\bar{X}) = \frac{\eta}{2} G_{\text{rand}} \bar{S}.$$

With

$$\eta = \frac{2\gamma_{M_{\text{gap}}}}{\lambda} = \frac{4\gamma_{M_{\text{gap}}} n^2}{M},$$

and using

$$\bar{G}\bar{S} = \lambda \bar{S}$$

by $\bar{S} \perp \mathbf{1}_n$, we obtain

$$f_{\theta_1}(\bar{X}) - \gamma_{M_{\text{gap}}} \bar{S} = \frac{\eta}{2}(G_{\text{rand}} - \bar{G})\bar{S}.$$

Next, we apply Lemma 3.3 with

$$\rho = \left(\frac{2\gamma_{M_{\text{gap}}} M_{\text{gap}}}{\log 2} \right)^2.$$

The stated width condition implies $M \geq Cn^2 \rho \log(2n/\delta)$, after absorbing the numerical factor $(2/\log 2)^2$ into the universal constant C. Hence, with probability at least $1 - \delta$, it holds that

$$\frac{\left\| G_{\text{rand}} - \bar{G} \right\|_2}{\lambda} \leq \frac{\log 2}{2\gamma_{M_{\text{gap}}} M_{\text{gap}}}.$$

On this event, we get

$$\left\| f_{\theta_1}(\bar{X}) - \gamma_{M_{\text{gap}}} \bar{S} \right\|_2 \leq \gamma_{M_{\text{gap}}} \frac{\left\| G_{\text{rand}} - \bar{G} \right\|_2}{\lambda} \left\| \bar{S} \right\|_2.$$

Since $\left\| \bar{S} \right\|_2 = \sqrt{n}$, we have

$$\frac{1}{n} \sum_{i=1}^{n} \left| \bar{s}_i f_{\theta_1}(\bar{x}_i) - \gamma_{M_{\text{gap}}} \right| = \frac{1}{n} \left\| f_{\theta_1}(\bar{X}) - \gamma_{M_{\text{gap}}} \bar{S} \right\|_1 \leq \gamma_{M_{\text{gap}}} \frac{\left\| G_{\text{rand}} - \bar{G} \right\|_2}{\lambda} \leq \frac{\log 2}{2M_{\text{gap}}}.$$

Let $\varphi(u) \doteq \log(1 + \exp(-u))$. Since $|\varphi'(u)| \leq 1$, it holds that φ is 1-Lipschitz. Also, for $y \in \{0, 1\}$ and $s = 2y - 1$, we have

$$\ell_{\log}(z, y) = \varphi(sz).$$

Therefore,

$$\mathcal{L}(\theta_1; \bar{\mathcal{T}}_{\text{cls}}) = \frac{1}{n} \sum_{i=1}^{n} \varphi(\bar{s}_i f_{\theta_1}(\bar{x}_i)) \leq \varphi(\gamma_{M_{\text{gap}}}) + \frac{\log 2}{2M_{\text{gap}}}.$$

By the definition

$$\gamma_{M_{\text{gap}}} = \log\left(\frac{2M_{\text{gap}}}{\log 2} \right),$$

we have

$$\varphi(\gamma_{M_{\text{gap}}}) = \log\big(1 + \exp\big(-\gamma_{M_{\text{gap}}}\big)\big) \leq \exp\big(-\gamma_{M_{\text{gap}}}\big) = \frac{\log 2}{2M_{\text{gap}}}.$$

Thus, we have

$$\mathcal{L}(\theta_1; \bar{\mathcal{T}}_{\text{cls}}) \leq \frac{\log 2}{M_{\text{gap}}}.$$

Since $\mathcal{L}(\theta_0; \bar{\mathcal{T}}_{\text{cls}}) = \log 2$, it follows that

$$\mathcal{G}^{(1)}_{\text{full}}(\theta_{\text{rand}}; \bar{\mathcal{T}}_{\text{cls}}) = \frac{\mathcal{L}(\theta_0; \bar{\mathcal{T}}_{\text{cls}}) - \mathcal{L}(\theta_1; \bar{\mathcal{T}}_{\text{cls}})}{\mathcal{L}(\theta_0; \bar{\mathcal{T}}_{\text{cls}})} \geq 1 - \frac{1}{M_{\text{gap}}}.$$

$\square$

B.6 Proof of Theorem 4.6

Proof. We have by Assumption 4.2

$$\mathcal{L}(\theta_1; \mathcal{T}) \le \mathcal{L}(\theta; \mathcal{T}) - \eta \langle g(\theta; \mathcal{T}), \widehat{g}_{\mathcal{B}_0}(\theta; \mathcal{T}) \rangle + \frac{\beta\eta^2}{2} \|\widehat{g}_{\mathcal{B}_0}(\theta; \mathcal{T})\|^2.$$

Then, by Assumption 4.1 and Theorem 2.1.5 of Nesterov [2013], we get

$$\mathbb{E}_{\mathcal{B}_0}[\mathcal{L}(\theta_1; \mathcal{T})] \le \mathcal{L}(\theta; \mathcal{T}) - \eta \|g(\theta; \mathcal{T})\|^2 + \frac{\beta\eta^2}{2} \mathbb{E}_{\mathcal{B}_0}\left[\|\widehat{g}_{\mathcal{B}_0}(\theta; \mathcal{T})\|^2\right]$$

$$\begin{aligned}
\mathcal{L}(\theta; \mathcal{T}) - \mathbb{E}_{\mathcal{B}_0}[\mathcal{L}(\theta_1; \mathcal{T})] &\ge \eta \|g(\theta; \mathcal{T})\|^2 - \frac{\beta\eta^2}{2} \mathbb{E}_{\mathcal{B}_0}\left[\|\widehat{g}_{\mathcal{B}_0}(\theta; \mathcal{T})\|^2\right] \\
&\ge \eta \|g(\theta; \mathcal{T})\|^2 - \beta\eta^2 \mathbb{E}_{\mathcal{B}_0}\left[\|\widehat{g}_{\mathcal{B}_0}(\theta; \mathcal{T})\|^2\right] \\
&= \eta \|g(\theta; \mathcal{T})\|^2 - \beta\eta^2 \frac{\|g(\theta; \mathcal{T})\|^2}{R(\theta; \mathcal{T})} \\
&= \eta \|g(\theta; \mathcal{T})\|^2 \left(1 - \frac{\beta\eta}{R(\theta; \mathcal{T})}\right).
\end{aligned}$$

Therefore, we further have

$$\begin{aligned}
\frac{\mathcal{L}(\theta; \mathcal{T}) - \mathbb{E}_{\mathcal{B}_0}[\mathcal{L}(\theta_1; \mathcal{T})]}{\mathcal{L}(\theta; \mathcal{T})} &\ge \eta \frac{\|g(\theta; \mathcal{T})\|^2}{\mathcal{L}(\theta; \mathcal{T})} \left(1 - \frac{\beta\eta}{R(\theta; \mathcal{T})}\right) \\
\mathcal{G}^{(1)}(\theta; \mathcal{T}) &\ge \eta S(\theta; \mathcal{T}) \left(1 - \frac{\beta\eta}{R(\theta; \mathcal{T})}\right).
\end{aligned}$$

Let $\eta = \alpha R(\theta; \mathcal{T})/\beta$. Since $0 < \eta < R(\theta; \mathcal{T})/\beta$, we then have $0 < \alpha < 1$. Plugging in η, we have

$$\begin{aligned}
\mathcal{G}^{(1)}(\theta; \mathcal{T}) &\ge \frac{\alpha(1-\alpha)}{\beta} S(\theta; \mathcal{T}) R(\theta; \mathcal{T}) \\
&= \frac{\alpha(1-\alpha)}{\beta} \mathrm{OR}(\theta; \mathcal{T}).
\end{aligned}$$

□

C Additional Experiment Details

C.1 Estimation Details

k-Step Gain Estimation. Let $\mathcal{T}_{\text{val}} \doteq (\mathcal{X}, \mathcal{Y}, P^{\text{val}}_{\mathcal{X}\mathcal{Y}})$ be a validation task and let θ be the checkpoint whose trainability we wish to estimate. We denote the validation set by

$$\mathcal{D}_{\text{val}} \doteq \{(x_i, y_i)\}_{i=1}^{N_{\text{val}}}, \qquad (x_i, y_i) \sim P^{\text{val}}_{\mathcal{X}\mathcal{Y}}.$$

We estimate the population loss $\mathcal{L}(\theta; \mathcal{T}_{\text{val}})$ using the empirical validation loss

$$\widehat{\mathcal{L}}_{\mathcal{D}_{\text{val}}}(\theta; \mathcal{T}_{\text{val}}) \doteq \frac{1}{N_{\text{val}}} \sum_{(x_i, y_i) \in \mathcal{D}_{\text{val}}} \ell(f_\theta(x_i), y_i).$$

In all experiments, we use $N_{\text{val}} = 10{,}000$.

Starting from $\theta_0 = \theta$, we run stochastic gradient descent for k steps,

$$\theta_{s+1} = \theta_s - \eta \widehat{g}_{\mathcal{B}_s}(\theta_s; \mathcal{T}_{\text{val}}), \qquad s = 0, \dots, k-1,$$

where each mini-batch $\mathcal{B}_s$ consists of $m = 4$ samples drawn uniformly with replacement from $\mathcal{D}_{\text{val}}$. We use $\eta = 10^{-3}$ for the gain-estimation rollouts. For each checkpoint and validation task, we repeat this procedure for $\mathcal{R} = 128$ independent rollouts and obtain terminal parameters $\left\{\theta_k^{(i)}\right\}_{i=1}^{\mathcal{R}}$. The expected post-update empirical loss is estimated as

$$\widehat{\mathcal{L}}_{\mathcal{R}}(\theta_k; \mathcal{T}_{\text{val}}) \doteq \frac{1}{\mathcal{R}} \sum_{i=1}^{\mathcal{R}} \widehat{\mathcal{L}}_{\mathcal{D}_{\text{val}}}\left(\theta_k^{(i)}; \mathcal{T}_{\text{val}}\right).$$

We then estimate the k-step gain by

$$\widehat{\mathcal{G}}^{(k)}(\theta;\mathcal{T}_{\text{val}}) \doteq \frac{\widehat{\mathcal{L}}_{\mathcal{D}_{\text{val}}}(\theta;\mathcal{T}_{\text{val}}) - \widehat{\mathcal{L}}_{\mathcal{R}}(\theta_k;\mathcal{T}_{\text{val}})}{\widehat{\mathcal{L}}_{\mathcal{D}_{\text{val}}}(\theta;\mathcal{T}_{\text{val}})}.$$

Optimization Readiness Estimation. We estimate the population gradient $g(\theta;\mathcal{T}_{\text{val}})$ using the gradient of the full validation set

$$\widehat{g}(\theta;\mathcal{T}_{\text{val}}) \doteq \nabla_\theta \widehat{\mathcal{L}}_{\mathcal{D}_{\text{val}}}(\theta;\mathcal{T}_{\text{val}}).$$

The empirical gradient strength is

$$\widehat{S}(\theta;\mathcal{T}_{\text{val}}) \doteq \frac{\|\widehat{g}(\theta;\mathcal{T}_{\text{val}})\|_2^2}{\widehat{\mathcal{L}}_{\mathcal{D}_{\text{val}}}(\theta;\mathcal{T}_{\text{val}})}.$$

To estimate gradient reliability, we independently sample $\mathcal{R} = 128$ mini-batches $\{\mathcal{B}_i\}_{i=1}^{\mathcal{R}}$ of size $m = 4$ from $\mathcal{D}_{\text{val}}$ with replacement and compute

$$\widehat{R}(\theta;\mathcal{T}_{\text{val}}) \doteq \frac{\|\widehat{g}(\theta;\mathcal{T}_{\text{val}})\|_2^2}{\frac{1}{\mathcal{R}}\sum_{i=1}^{\mathcal{R}} \|\widehat{g}_{\mathcal{B}_i}(\theta;\mathcal{T}_{\text{val}})\|_2^2}.$$

The estimated optimization readiness is then

$$\widehat{\text{OR}}(\theta;\mathcal{T}_{\text{val}}) \doteq \widehat{S}(\theta;\mathcal{T}_{\text{val}})\widehat{R}(\theta;\mathcal{T}_{\text{val}}).$$

Baseline Metric Estimation. Let X_{val} denote the matrix of validation inputs, where the i-th row is $x_i^\top$. For both SCR and P-MNIST, we compute the representation effective rank and 99% energy rank as

$$\text{EffRank}(\Phi_\theta(X_{\text{val}})), \qquad \text{EnRank}_{0.99}(\Phi_\theta(X_{\text{val}})).$$

We also compute the active-neuron fraction on the full validation set using threshold $\tau_{\text{act}} = 0.1$.

For SCR, the network has a manageable number of parameters, so we compute the eNTK rank and exact Hessian rank directly:

$$\text{EnRank}_{0.99}\Big(\widehat{K}(X_{\text{val}};\theta)\Big), \qquad \text{EnRank}_{0.99}\Big(\nabla_\theta^2 \widehat{\mathcal{L}}_{\mathcal{D}_{\text{val}}}(\theta;\mathcal{T}_{\text{val}})\Big).$$

For P-MNIST, we omit the eNTK rank because $f_\theta(x)$ is vector-valued, so $\nabla_\theta f_\theta(x)$ is a Jacobian rather than a vector and the resulting empirical kernel is not directly comparable to the scalar-output case.

For the Hessian-rank metric in P-MNIST, we follow the Gram-matrix approximation used by Lewandowski et al. [2024]. Specifically, we sample $b = 128$ input-label pairs from $\mathcal{D}_{\text{val}}$ and form

$$G \doteq [\nabla_\theta \ell(f_\theta(x_1), y_1) \quad \cdots \quad \nabla_\theta \ell(f_\theta(x_b), y_b)].$$

We then use

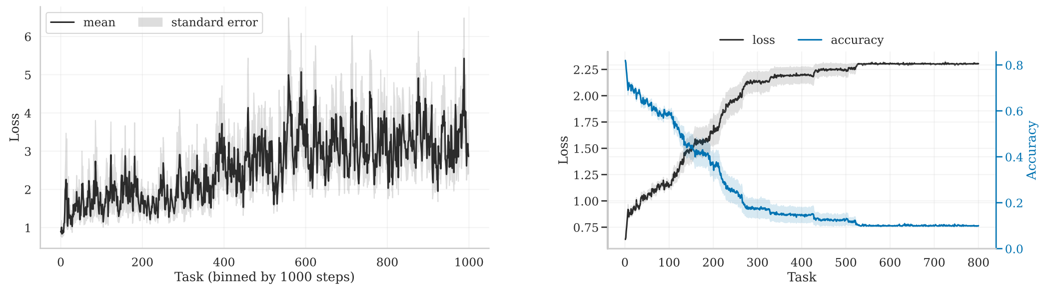
$$\text{EnRank}_{0.99}(G^\top G)$$

as the approximated 99% energy rank of the Hessian.

C.2 Compute Resources

Experiments were run on a shared Slurm cluster. Since jobs were scheduled opportunistically, the exact hardware allocation varied across runs. SCR experiments were faster on CPU and took less than 3 hours per seed. P-MNIST experiments benefited from GPU training and took approximately 10.5 hours per seed on an NVIDIA RTX 4000 GPU with 8GB of GPU memory.

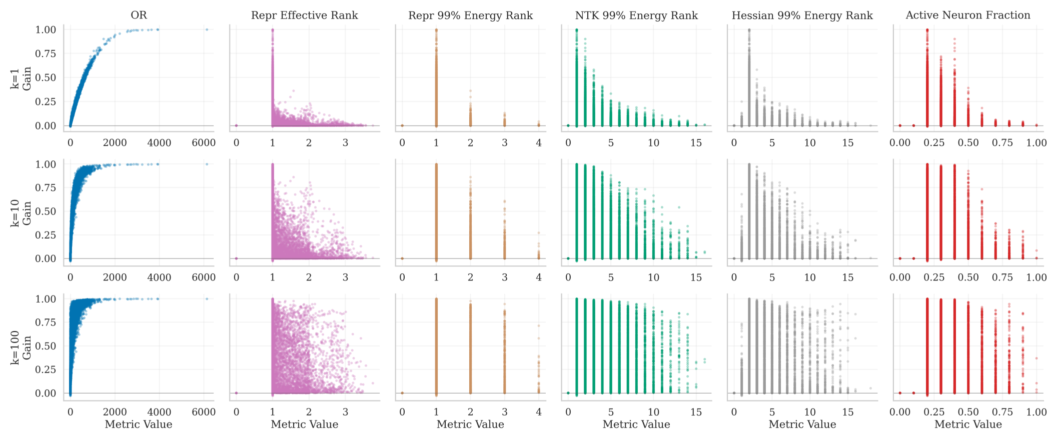
C.3 Figures



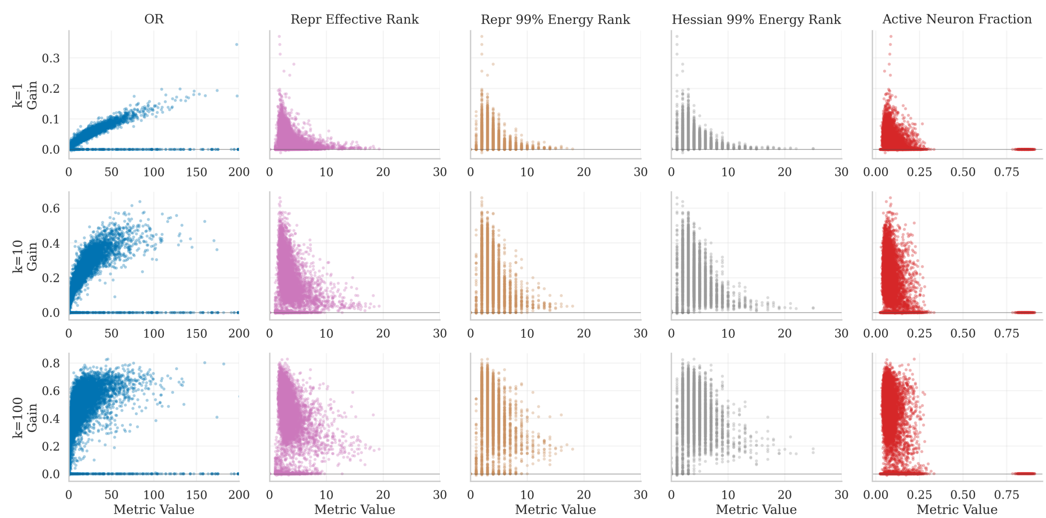
(a) Slowly-Changing Regression

(b) Permuted MNIST

Figure 1: Learning curves of Slowly-Changing Regression and Permuted MNIST.

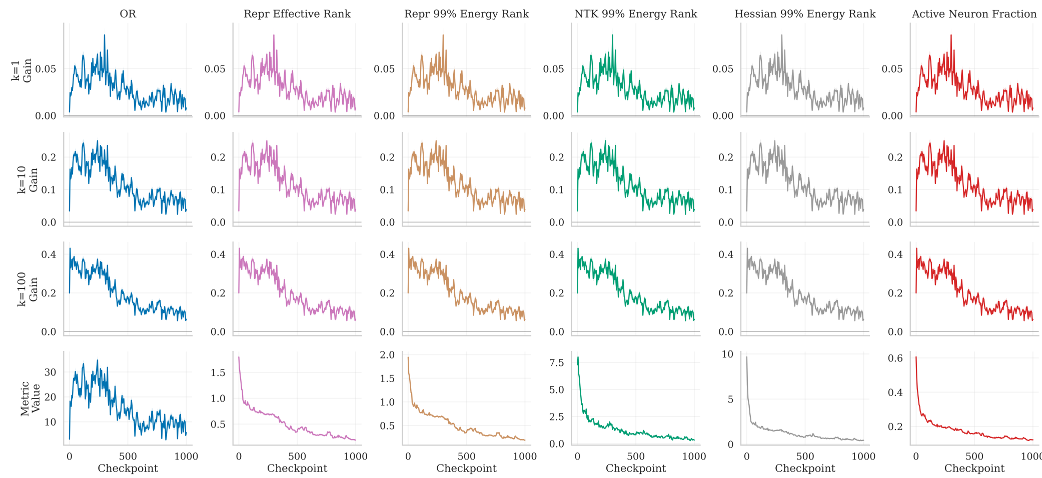


(a) Slowly-Changing Regression. Results aggregated from 20 training runs and 30 validation tasks.

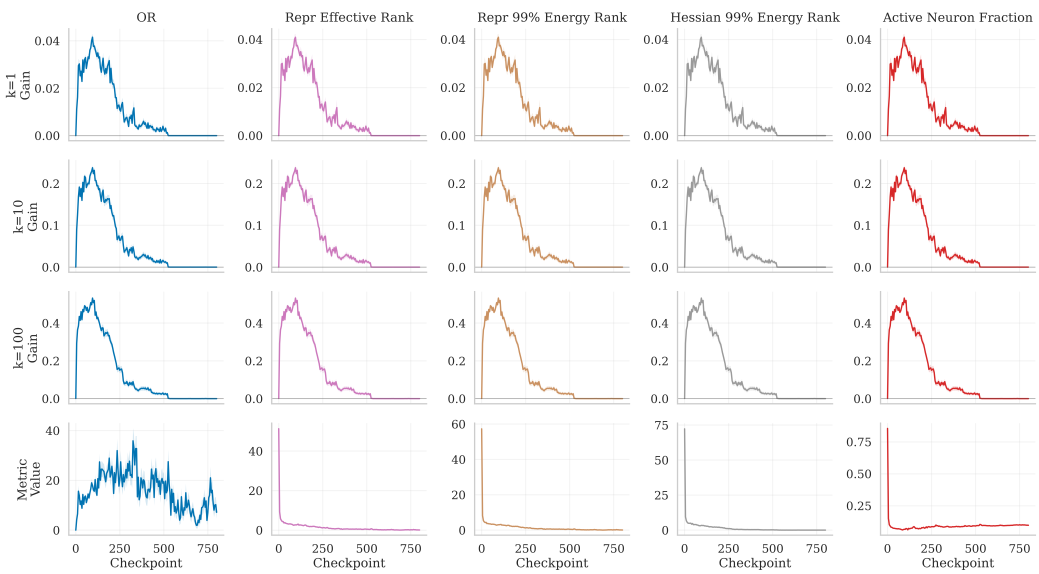


(b) Permuted MNIST. Results aggregated from 20 training runs and 10 validation tasks.

Figure 2: 1-, 10-, and 100-step gain vs. plasticity metric values for Slowly-Changing Regression and Permuted MNIST (extreme metric values clipped to account for outliers).

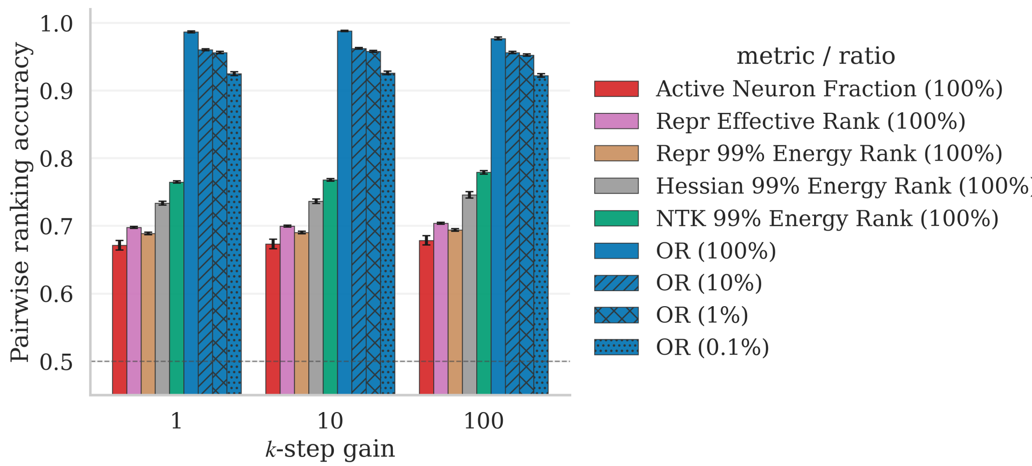


(a) Slowly-Changing Regression. Results aggregated from 20 training runs and 30 validation tasks.

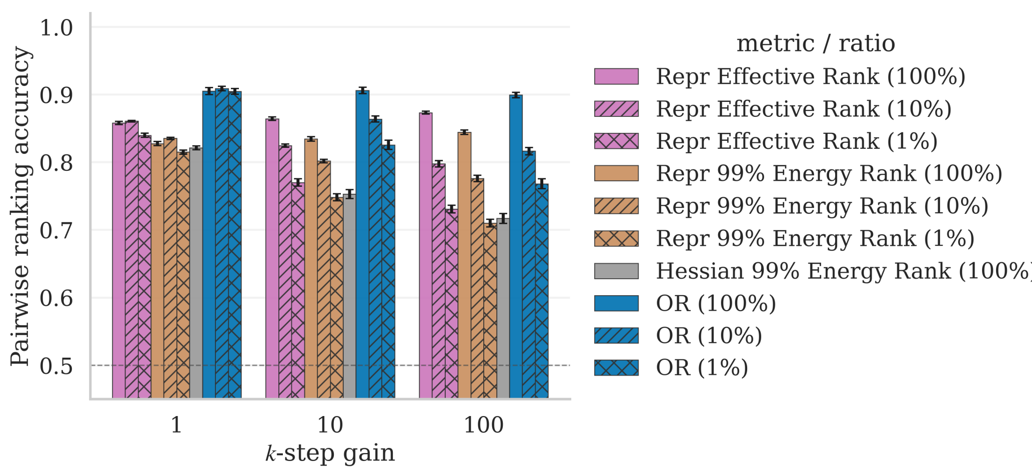


(b) Permuted MNIST. Results aggregated from 20 training runs and 10 validation tasks.

Figure 3: 1-, 10-, and 100-step gain and plasticity metric values against checkpoints for Slowly-Changing Regression and Permuted MNIST (extreme metric values clipped to account for outliers). Shaded areas denote standard errors.



(a) Slowly-Changing Regression. Results aggregated from 20 training runs and 30 validation tasks.



(b) Permuted MNIST. Results aggregated from 20 training runs and 10 validation tasks. We omit the active neuron fraction because its accuracy is all under 50%.

Figure 4: Subsampling ablation study results for Slowly-Changing Regression and Permuted MNIST. The error bars denote standard errors.